

Slide Deck 1 - Math 200

Multivariate functions

A two variable function maps ordered pairs to a unique real number

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

and $z = f(x, y)$.

Domain

The domain is defined as

$$\text{dom}(f) = \{(x, y) : f(x, y) \in \mathbb{R}\}$$

Range

The range is defined as

$$\text{rng}(f) = \{z : \exists(x, y) \in \text{dom}(f) \text{ with } f(x, y) \in z\}$$

Distance in $\mathbb{R}^n$

In $\mathbb{R}^n$, the distance is given as

$$\text{dist}(x, a) = \sqrt{(x_1 - a_1)^2 + (x_2 - a_2)^2 + \cdots + (x_n - a_n)^2}$$

Basic Transformations

$g(x, y) = f(x, y) + K$	The graph of g is the graph of f moved up by K units.
$h(x, y) = -f(x, y)$	The graph of h is reflected through the xy -plane.
$k(x, y) = Lf(x, y)$	The graph of k is the graph of f stretched vertically by a factor of L
$m(x, y) = f(x - a, y - b)$	The graph of m is the graph of f shifted a units in the x -direction and b units in the y -direction

Linear Functions

A linear function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function of the form

$$f(x, y) = m_x x + m_y y + c$$

In general, a linear function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function of the form

$$f(x, y) = m_1 x_1 + m_2 x_2 + \cdots + m_n x_n + c$$

Limit for $f : \mathbb{R}^1 \rightarrow \mathbb{R}$

Recall that

$$\lim_{x \rightarrow a} f(x) = L$$

means that,

- The limit of f as x approaches a is L .
- For every $\varepsilon > 0$ there exists $\delta > 0$ such that if $|x - a| < \delta$ with $x \neq a$, then $|f(x) - L| < \varepsilon$.

For $x \in \mathbb{R}^1, a \in \mathbb{R}^1, f : \mathbb{R}^1 \rightarrow \mathbb{R}$ the $\lim_{x \rightarrow a} f(x) = L$ means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $\text{dist}(x, a) < \delta$ with $x \neq a$ then $|f(x) - L| < \varepsilon$.

Limit for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

For $x \in \mathbb{R}^2, a \in \mathbb{R}^2, f : \mathbb{R}^2 \rightarrow \mathbb{R}$ the $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $\text{dist}((x, y), (a, b)) < \delta$ with $x \neq a$ then $|f(x) - L| < \varepsilon$.

Limit for $f : \mathbb{R}^n \rightarrow \mathbb{R}$

For $x \in \mathbb{R}^n, a \in \mathbb{R}^n, f : \mathbb{R}^n \rightarrow \mathbb{R}$ the $\lim_{x \rightarrow a} f(x) = L$ means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $\text{dist}((x), (a)) < \delta$ with $x \neq a$ then $|f(x) - L| < \varepsilon$.

Continuity

For $(x, y) \in \mathbb{R}^2, (a, b) \in \mathbb{R}^2, f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the function f is continuous at (a, b) iff

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

For $x \in \mathbb{R}^n, a \in \mathbb{R}^n, f : \mathbb{R}^n \rightarrow \mathbb{R}$, the function f is continuous at a iff

$$\lim_{x \rightarrow a} f(x) = f(a)$$

In $\mathbb{R}^n$, this implies three things:

1. $f(x)$ exists at and near a
2. $\lim_{x \rightarrow a} f(x)$ is well defined
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Limit Laws

Theorem (Sums and Products). Suppose there are two functions of one variable, $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a and $h : \mathbb{R} \rightarrow \mathbb{R}$ and is continuous at b . Then $f(x, y) = g(x) + h(y)$ is continuous at (a, b) and $f(x, y) = g(x)h(y)$ is continuous at (a, b) .

Theorem (Compositions). Suppose $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous at (a, b) and $h : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at $z = g(a, b)$. Then $f(x, y) = h(g(x, y))$ is continuous at (a, b) .

Derivative for $f : \mathbb{R}^1 \rightarrow \mathbb{R}$

Recall that

$$f'(x) = \frac{d}{dx}f(x) = \left. \frac{df}{dx} \right|_x = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Tangent Plane

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, we don't have a tangent line, we have a tangent plane.

Partial Derivatives

Definition. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. The partial derivative of f with respect to x is given by

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

provided this limit exists.

The partial derivative of f with respect to y at (a, b) is given by

$$\frac{\partial f}{\partial y}(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h}$$

provided this limit exists.

Differentiability

Definition. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. f is *partially* differentiable if *all* partial derivatives exist. That is:

$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \dots \right) \text{ exist}$$

f is *fully* differentiable when

$$\frac{\partial f}{\partial x}(a, b), \frac{\partial f}{\partial y}(a, b), \text{ etc. exist.}$$

Example

Consider $f(x, y) = x^{1/4}y^{1/8}$. See if f is partially differentiable at $(0, 0)$.

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0 + h, 0) - f(0, 0)}{h} \rightarrow 0.$$

Similarly,

$$\frac{\partial f}{\partial y}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} \rightarrow 0.$$

Therefore f is partially differentiable at $(0, 0)$.

Now, let $a, b \neq 0$. Therefore

$$f_x(a, b) = \dots = \frac{b^{1/8}}{4a^{3/4}} \xrightarrow{\lim_{(a,b) \rightarrow (0,0)}} \infty$$

and

$$f_y(a, b) = \dots = \frac{a^{1/4}}{8b^{7/8}} \rightarrow 0$$

Therefore f is not a continuous function.

Differentiability $\implies$ continuity

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is partially differentiable at (a, b) then

$$\lim_{x \rightarrow a} f(x, b) = f(a, b)$$

and

$$\lim_{y \rightarrow b} f(a, y) = f(a, b).$$

Higher order partials

Definition. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. The second order partial derivative of f at with respect to x at y is given by

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$$

provided the limits exist.

Example

Consider $f(r, \theta, z) = r \sin(\theta) - z^2$. Find all second order partial derivatives of f .

In other words, find

$$\frac{\partial^2}{\partial r \partial r}, \frac{\partial^2}{\partial r \partial \theta}, \frac{\partial^2}{\partial r \partial z}, \dots, \frac{\partial^2}{\partial z \partial z}$$

Consider the first order partials first.

$$\frac{\partial f}{\partial r} = \sin \theta, \quad \frac{\partial f}{\partial \theta} = r \cos \theta, \quad \frac{\partial f}{\partial z} = -2z$$

Carry on with the second order partials.

$$\begin{aligned} \frac{\partial^2 f}{\partial r \partial r} &= \frac{\partial}{\partial r}(\sin \theta) = 0, & \frac{\partial^2 f}{\partial \theta \partial r} &= \frac{\partial}{\partial \theta}(\sin \theta) = \cos \theta, & \frac{\partial^2 f}{\partial z \partial r} &= \frac{\partial}{\partial z}(\sin \theta) = 0 \\ \frac{\partial^2 f}{\partial r \partial \theta} &= \frac{\partial}{\partial r}(r \cos \theta) = \cos \theta, & \frac{\partial^2 f}{\partial \theta \partial \theta} &= \frac{\partial}{\partial \theta}(r \cos \theta) = -r \sin \theta, & \frac{\partial^2 f}{\partial z \partial \theta} &= \frac{\partial}{\partial z}(r \cos \theta) = 0 \\ \frac{\partial^2 f}{\partial r \partial z} &= 0, & \frac{\partial^2 f}{\partial \theta \partial z} &= 0, & \frac{\partial^2 f}{\partial z \partial z} &= -2 \end{aligned}$$

Clairaut's Theorem

Theorem. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Suppose that $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are fully differentiable (i.e. all 2nd order partials of f are continuous) then

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$